Consumer Electronics Sales Analytics & Customer Insights System

Tools: *Python • SQL • Power BI • Excel*

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Business Problem

The dataset represents comprehensive sales transactions in the consumer electronics domain, encompassing customers, products, orders, and reviews across multiple touchpoints.

Revenue Analysis

Understanding which products and categories generate the highest revenue streams

Customer Insights

Identifying top customers contributing to overall sales performance

Trend Analysis

Analyzing sales trends over time to identify patterns and opportunities

Product Performance

Evaluating product performance based on ratings and pricing strategies

Objective: *To analyze sales and customer data in order to extract insights that support data-driven decision-making*

Dataset Overview

The dataset represents a comprehensive sales system in the consumer electronics domain, consisting of multiple related tables that capture the complete customer journey from purchase to post-purchase satisfaction.

Customers

50,000+ *customers with demographic information including age, gender, and location data*

Orders

200,000+ *transactions capturing order dates, amounts, and customer relationships*

Order Items

10,000+ *line items detailing each product within an order with pricing and quantities*

Products

1,000+ *products including category, price, and brand relationships*

Brands

23+ *brands representing product manufacturers and their origin countries*

Reviews

100,000+ *customer ratings capturing satisfaction scores for products purchased*

Monthly Revenue Trends

The Big Jump

June saw a massive surge, climbing from $142.14M in May to $159.17M

The Yearly Peak

July was the most successful month of the entire year, reaching a record high of $214.20M

Sustained Momentum

August maintained this strength, staying high at $209.87M

Strategic Insight: *The data suggests that the "Summer" period is a critical sales window for this company. After a slow spring (April being the lowest point), the business regained its footing in June and hit its maximum potential in July.*

Product Performance

Financial Overview

$2.3B	50K	1.5M	$11.42K
Total Revenue	Customer Base	Products Sold	Average Order Value
Driven by a portfolio of 23 global brands	*Serving 50,000 customers*	*Impressive transaction volume*	*Indicating high-ticket retail model*

Market & Brand Leaders

- ***Top Market:*** *The USA is the primary revenue generator, supported by strong footprints in China, South Korea, and Switzerland*
- ***Dominant Brands:*** *Microsoft leads with $135.19M in revenue, followed closely by Lenovo ($131.73M) and Acer ($128.92M)*
- ***Star Product:*** *The "Statement Air" is the top-performing individual product, contributing $9.24M*

Growth Trends

- ***Long-term Trend:*** *The business hit a massive historical peak in 2022 ($327.14M). While 2023-2024 saw a slight decline, 2025 shows a recovery trend moving back toward $275M*
- ***Seasonal Cycle:*** *Within a single year, July is the "Golden Month" ($214.20M), marking a sharp recovery from a yearly low in April ($139.65M)*

Product Category Strength

- ***Core Revenue Drivers:*** *Smartphones ($191.65M), Cameras, and Printers are the most profitable categories*
- ***Growth Opportunities:*** *Smartwatches and Speakers currently sit at the bottom of the revenue list, suggesting room for better marketing or inventory adjustments*

Customer Insights

Customer Overview

Total Customers
50,000 consistent customer base

Average Age
41.46 years, mature customer base

Top Customer
David Johnson by revenue

Top City
West Michael by revenue

Top Customers by Revenue

01

David Johnson
Leader contributing $1.20M in revenue

02

Michael Johnson
Runner-up with $1.01M

03

High-Value Group
Several others clustered in $900K - $960K range

Demographic Distribution

Total Revenue by Gender

The pie chart shows an almost perfect 50/50 split, indicating the products and brand appeal equally to everyone:

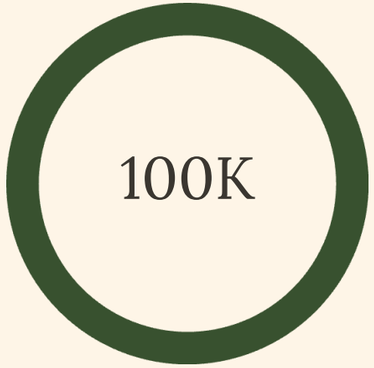
- ***Male:*** *50.32%*
- ***Female:*** *49.68%*

Geographic Distribution

The "Total Revenue by City" map shows global hotspots with significant activity in North America (USA) and Western Europe, with large blue "heat" circles appearing over parts of Asia, confirming strong performance in China, South Korea, and Japan.

Rating Insights

Satisfaction Benchmarks



Total Reviews

Very large and reliable sample size

Average Rating

Perfectly neutral score

Top Performers by Category

- ***Top Categories:*** *Routers and Keyboards receive the highest satisfaction scores from users*
- ***Top Brands:*** *Nikon and Apple are the most trusted brands by customers in terms of quality and experience*
- ***Top Countries:*** *Customers in China and the USA provide the highest ratings for the products they receive*

Average Rating Trend (Monthly)

The line chart shows how customer satisfaction fluctuates throughout the year:

- ***Peak Satisfaction:*** *May saw the highest average rating of 3.039*
- ***The Slump:*** *There was a noticeable dip in June (2.992), right after the yearly peak*
- ***Consistency:*** *For most of the year, the rating stays extremely close to the 3.0 mark, showing very stable (but average) customer sentiment*

Top 10 Highest Rated Products

- ***Leaders:*** *Agreement Pro and Sister Air are tied for the #1 spot with a rating of 3.41*
- ***High Performers:*** *Products like "Purpose Air" (3.39) and "Become Max" (3.37) also perform well above the 3.00 average*
- ***Consistency:*** *Even at the bottom of the "Top 10" (Local Pro), the rating is still a solid 3.32*

Key Findings

Revenue Concentration by Product

A high percentage of total revenue is concentrated in specific categories. Smartphones lead significantly ($191.65M), followed by Cameras and Printers, indicating the business's heavy reliance on these core tech sectors.

Top-Tier Customer Revenue Influence

A very small percentage of customers generates a disproportionate amount of revenue. With individual customers like David Johnson contributing over $1.20M, the business relies heavily on high-value "VIP" relationships.

Strategic Revenue vs. Rating Divergence

There is a notable difference between top sales and top satisfaction. While Microsoft leads in total revenue ($135.19M), Nikon and Apple dominate customer satisfaction scores, suggesting that the highest-earning brands are not necessarily the ones customers "love" the most.

Historical Peak and Current Recovery

After reaching a historic high in 2022 ($327.14M), the company experienced a sharp correction. However, 2025 data shows a strong recovery trend, with revenue climbing back toward the $275M mark.

Perfect Demographic Balance

The business appeals equally to both male and female customers, showing a remarkably balanced split of 50.32% Male vs. 49.68% Female, which simplifies target marketing and confirms broad-spectrum product appeal.

Total Amount Transaction Outliers

The system shows significant outliers in total transaction amounts. With an average order value of $11.42K, specific transactions and high-net-worth clients contribute amounts that far exceed this benchmark, pointing to a strong luxury or corporate sector presence.

Strategic Recommendations

1 Target the "Revenue-Satisfaction" Gap

The Data: *Your highest revenue brand is Microsoft ($135.19M), but the highest-rated brands are Nikon and Apple.* ***Recommendation:*** *Conduct a deep-dive analysis into Microsoft product reviews to identify common pain points. Improving the satisfaction of your highest-selling brand could prevent future churn to better-rated competitors.*

2 Premium Loyalty Program for "Elite" Customers

The Data: *A single customer, David Johnson, generated $1.20M in revenue. The top 7 customers all exceed $740K each.* ***Recommendation:*** *Implement a "Platinum Tier" loyalty program specifically for these high-net-worth individuals. Offer dedicated account managers, early access to new releases (like the Statement Air), and bespoke technical support to secure this critical revenue stream.*

3 Reverse the Q1–Q2 Seasonal Slump

The Data: *Monthly trends show a significant revenue drop from January ($209M) down to April ($139M).* ***Recommendation:*** *Launch aggressive "Spring Tech Sales" or bundle offers in March and April. Focus on categories that are currently lower in revenue, like Smartwatches or Tablets, to boost transaction volume during this historical low period.*

4 Expand "Best-in-Class" Categories

The Data: *Routers and Keyboards have the highest average ratings, but they rank in the middle or bottom of total revenue charts.* ***Recommendation:*** *Increase marketing spend for Routers and Keyboards. Since customers already love these products, higher visibility (SEO/Ads) is likely to result in high conversion rates and positive word-of-mouth.*

5 Geographical Market Expansion

The Data: *The USA is your top country by revenue, but your global heatmap shows significant untapped potential in the South Pacific and parts of South America.* ***Recommendation:*** *Explore localized marketing campaigns in Taiwan and the Netherlands, which are already identified as key markets but may have room for growth to match USA levels.*

6 Data Schema Optimization (Technical)

The Data: *The schema shows a Reviews table linked to Products.* ***Recommendation:*** *Bridge your "Customers" and "Reviews" data to perform Sentiment Analysis by Demographics. Understanding if specific age groups (Current Average: 41.46) prefer certain brands will allow for more "Student-style" or age-appropriate personalized marketing.*

Thank You

Questions & Answers